



(Autonomous College Affiliated to University of Mumbai)

Department of Information Technology

Internship Report

On

"AI & Cloud Research, Generative AI and Cloud Computing"

At

INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI)

Faculty Mentor

Prof. Pallavi Kharat

Name of the faculty

From 01/11/2025 to 30/04/2026

Submitted by

Riya Sunil Kharade

Roll No.: 24101C2001

Semester VI

Department of Information Technology

Certificate

This is to certify that the internship work presented in this report entitled “Topic/ Area/ Project name” during the period dd/mm/yyyy to dd/mm/yyyy was carried out by student name (Roll No./Div) of Department of Information Technology at Vidyalankar Institute of Technology, Mumbai for partial fulfillment of credits of semester VI. The internship work that has been carried out is to the satisfaction of the department.

Place: Mumbai

Date:11/05/2026

Name & Signature of faculty Mentor

Signature of Head, Dept. of Information Technology

Completion Certificate

INTELLIGENT SYSTEMS PRIVATE LIMITED
CIN: U62090KA2025PTC210049
<https://www.enterprisesi.co>



Certificate of completion

This is to certify that Riya Sunil Kharade, pursuing B.Tech. in Information Technology from Vidyalankar Institute of Technology, Mumbai has successfully completed 6 months of internship from **1st November 2025 to 30th April 2026** with **INTELLIGENT SYSTEMS PRIVATE LIMITED** ("EnterpriseSI"), an AI/Generative AI Acceleration Platform startup based in Bangalore.

During the internship period, Riya Sunil Kharade worked on Generative AI, AI Agents, RAG and related research & code development for the Artificial General Intelligence (AGI) Playground platform and consulting work that EnterpriseSI offers to its customers.

We wish her all the best for her future endeavors.

Abhiroop Roy
CEO & Founder
EnterpriseSI

Date: 11.05.2026

Acknowledgement

I sincerely express my gratitude to INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) for providing me with the opportunity to work as an AI & Cloud Research Intern. This internship provided valuable exposure to Artificial Intelligence, Generative AI, Cloud Computing, AI Agents, and modern software development practices.

I would like to thank Mr. Abhiroop Roy (CEO & Founder) for offering me this opportunity and for his guidance throughout the internship. I am also thankful to my mentors Mr. Sachitra and Mr. Tushar for assigning weekly learning tasks, reviewing my work, sharing valuable resources, and helping me understand emerging AI technologies.

I sincerely thank Prof. Pallavi Kharat, Faculty Mentor, Department of Information Technology, Vidyalkar Institute of Technology, for her continuous support and guidance throughout the internship period.

Finally, I would like to thank my parents, teachers, and friends for their encouragement and motivation, which helped me successfully complete this internship.

Abstract

This report presents the work carried out during my six-month internship as an AI & Cloud Research Intern at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) from 1 November 2025 to 30 April 2026. The internship focused on developing knowledge and practical skills in Artificial Intelligence, Generative AI, Cloud Computing, AI Agents, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and workflow automation.

Throughout the internship, I completed weekly research-based assignments, prepared technical documentation, and developed small proof-of-concept applications using technologies such as Node.js, React.js, and n8n. I also gained hands-on exposure to AWS services, including EC2, S3, IAM, and DynamoDB, and explored cloud architecture concepts for AI applications.

The internship emphasized independent learning through research, technical documentation, experimentation with AI tools, and periodic mentor reviews. This experience strengthened my technical knowledge, problem-solving abilities, research skills, and understanding of modern AI and cloud technologies while preparing me for future roles in software engineering and AI development.

Table of Contents

Acknowledgement		
Abstract		
Chapter 1	Introduction	
1.1	Overview of the Internship.....	
1.2	Objective of the Internship.....	
1.3	Importance of the Internship in Engineering Education.....	
Chapter 2	Company/Organization Profile	
2.1	History and Background of the Organization.....	
2.2	Mission and Vision.....	
2.3	Organizational Structure.....	
2.4	Services/Products Offered.....	
2.5	Overview of Departments.....	
Chapter 3	Internship Role and Responsibilities	
3.1	Internship Position/Title.....	
3.2	Department Assigned.....	
3.3	Detailed Roles and Responsibilities.....	
3.4	Project(s) Assigned.....	
3.5	Technologies, Tools, and Software Used.....	
3.6	Work Environment and Team Dynamics.....	
Chapter 4	Work and Learning Outcomes	
4.1	Detailed Description of Work Done.....	
4.2	Key Learning in Technical and Soft Skills.....	

4.3	Challenges Faced and Solutions Applied.....	
4.4	Contributions to the Organization.....	
4.5	Application of Theoretical Knowledge in Practical Work.....	
Chapter 5	Industrial Exposure and Skills Development	
5.1	Understanding Industry Standards and Practices.....	
5.2	Interaction with Professionals and Mentors.....	
5.3	Enhancement of Professional Skills (communication, teamwork, time management)	
5.4	Networking and Career Growth Insights.....	
Chapter 6	Challenges and Recommendations	
6.1	Major Challenges Encountered During Internship.....	
6.2	How They Were Addressed.....	
6.3	Suggestions for Improvement in Future Internships.....	
Chapter 7	Conclusion	
7.1	Summary of Internship Experience.....	
7.2	Key Takeaways and Personal Growth.....	
7.3	How the Internship Influences Future Career Goals.....	
7.4	Overall Evaluation of the Internship Program.....	
Chapter 8	References	
	Citations for any sources, articles, or guidelines followed during the internship	
Appendix		
A	Project Documentation (if applicable).....	
B	Sample Code/Design/Reports (if applicable).....	
C	Additional Certificates or Achievements.....	

CHAPTER 1

Introduction

1.1 Overview of the Internship

During my six-month internship at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI), I worked as an AI & Cloud Research Intern from 1 November 2025 to 30 April 2026. The internship was conducted in a mostly remote mode, where weekly tasks were assigned through Google Meet sessions and communication was maintained via a WhatsApp group. The completed work was submitted through Google Drive, while I maintained a personal GitHub repository to organize my learning and project work.

The internship focused on emerging technologies such as Artificial Intelligence (AI), Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Cloud Computing, and workflow automation using n8n. Throughout the internship, I completed research-based assignments, prepared technical documentation, explored cloud technologies, and developed small proof-of-concept applications using Node.js and React.js.

I also gained practical exposure to Amazon Web Services (AWS), including EC2, S3, IAM, and DynamoDB, and learned how cloud services support modern AI applications. The internship emphasized independent learning, technical research, documentation, and continuous improvement through mentor feedback and periodic presentations.

1.2 Objectives of the Internship

The primary objectives of the internship were:

- To gain practical knowledge of Artificial Intelligence, Generative AI, and Cloud Computing.
- To understand the working principles of Large Language Models (LLMs), AI Agents, and Retrieval-Augmented Generation (RAG).
- To develop hands-on experience with AWS services such as EC2, S3, IAM, and DynamoDB.
- To learn workflow automation using n8n for business applications.
- To strengthen programming skills using Node.js and React.js through small demonstration projects.
- To improve research, documentation, analytical thinking, and problem-solving abilities.
- To understand current industry practices and emerging AI technologies.

1.3 Importance of the Internship in Engineering Education

An internship plays a significant role in bridging the gap between academic learning and industry practices. It provides engineering students with practical exposure to real-world technologies, development processes, and professional work environments.

Through this internship, I was able to apply the concepts learned during my academic studies to practical tasks involving Artificial Intelligence and Cloud Computing. The experience enhanced my technical knowledge, research capabilities, communication skills, and confidence in working with modern software technologies. It also improved my ability to learn independently, collaborate with mentors and peers, and adapt to rapidly evolving technologies.

Overall, the internship strengthened my understanding of industry expectations and motivated me to pursue a career in Artificial Intelligence, Cloud Computing, and software development.

CHAPTER 2

Company / Organization Profile

2.1 History and Background of the Organization

INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) is an Artificial Intelligence and Generative AI acceleration platform startup based in Bengaluru, Karnataka, India. The company specializes in helping organizations adopt modern AI technologies by providing consulting services, AI solutions, research, and hands-on learning platforms.

EnterpriseSI focuses on accelerating AI adoption through innovative technologies such as Generative AI, Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Cloud Computing, and Artificial General Intelligence (AGI). During my internship, I contributed to research and code development for the AGI Playground platform and participated in AI-related research activities aligned with the company's consulting services.

The company promotes continuous learning and encourages interns to explore emerging technologies through research, practical assignments, and proof-of-concept implementations. This approach helps students gain exposure to current industry trends while developing both technical and analytical skills.

2.2 Mission and Vision

Mission

To accelerate the adoption of Artificial Intelligence and Cloud technologies by delivering innovative AI solutions, research, consulting services, and practical learning experiences that help organizations solve real-world business challenges.

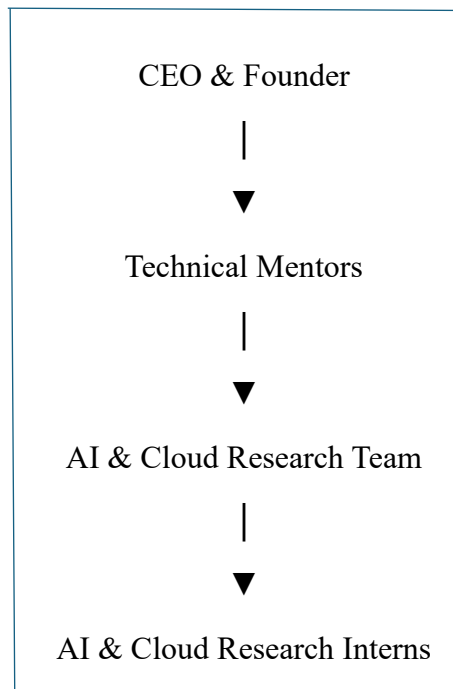
Vision

To become a leading AI and Generative AI innovation company by empowering businesses and individuals through advanced AI technologies, cloud platforms, and next-generation intelligent systems.

2.3 Organizational Structure

EnterpriseSI follows a collaborative organizational structure where experienced professionals, technical mentors, and interns work together on AI research and development activities.

A simplified organizational structure is shown below:



During my internship, I worked under the guidance of Mr. Abhiroop Roy (CEO & Founder) and received technical mentorship from Mr. Sachitra and Mr. Tushar, who assigned weekly tasks, reviewed submissions, and provided guidance during Google Meet sessions.

2.4 Services / Products Offered

EnterpriseSI provides solutions and services in the following domains:

- Artificial Intelligence (AI) Consulting
- Generative AI Solutions
- Large Language Model (LLM) Applications
- AI Agent Development
- Retrieval-Augmented Generation (RAG)
- Artificial General Intelligence (AGI) Research
- Cloud Computing Solutions
- AI and Cloud Training Programs
- Enterprise AI Consulting
- Proof-of-Concept (PoC) Development
- AI Application Modernization

One of the company's major initiatives is the AGI Playground, which serves as a platform for researching, experimenting with, and developing intelligent AI systems and modern Generative AI applications.

2.5 Overview of Departments

EnterpriseSI operates across multiple functional areas that work together to develop AI-driven solutions. The major departments include:

AI Research & Development

Focuses on research in Generative AI, LLMs, AI Agents, and emerging AI technologies.

Cloud Solutions

Works with cloud platforms such as AWS and other cloud technologies to build scalable AI applications.

AI Consulting

Provides AI consulting services and enterprise solutions for organizations adopting Artificial Intelligence.

Product Development

Develops AI platforms, proof-of-concept applications, and intelligent software solutions, including the AGI Playground.

Training and Knowledge Development

Conducts technical learning initiatives, research activities, documentation, and skill development programs for interns and professionals.

CHAPTER 3

Internship Role and Responsibilities

3.1 Internship Position / Title

I worked as an AI & Cloud Research Intern at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) from 1 November 2025 to 30 April 2026. The internship was primarily research-oriented and focused on Artificial Intelligence, Generative AI, Cloud Computing, and emerging AI technologies. My role involved studying new technologies, preparing technical documentation, exploring cloud platforms, and developing small proof-of-concept applications.

The internship provided practical exposure to modern AI concepts and encouraged independent learning through weekly assignments and mentor guidance.

3.2 Department Assigned

I was assigned to the AI & Cloud Research Department, where the primary focus was on researching and exploring Artificial Intelligence, Generative AI, AI Agents, Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Cloud Computing technologies.

The department promoted continuous learning through weekly research tasks, technical documentation, and practical experimentation. Mentors guided interns by assigning new topics, reviewing completed work, and discussing the concepts during online sessions.

3.3 Detailed Roles and Responsibilities

During the internship, I performed the following responsibilities:

- Conducted research on Artificial Intelligence, Generative AI, and Cloud Computing technologies.
- Studied the working principles of Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), and Artificial General Intelligence (AGI).
- Prepared detailed technical documentation based on weekly research topics.
- Built automation workflows using n8n to understand workflow automation concepts.
- Learned and explored AWS cloud services, including EC2, S3, IAM, and DynamoDB, to understand cloud infrastructure and AI deployment.
- Developed small demonstration projects using Node.js and React.js.

- Explored technologies such as Langflow, Model Context Protocol (MCP), Ollama, and Groq LLM.
- Participated in Google Meet sessions where mentors explained new topics, reviewed completed tasks, and provided feedback.
- Submitted weekly assignments and documentation through Google Drive.
- Maintained a personal GitHub repository to organize project files, code, documentation, and learning resources.

3.4 Project(s) Assigned

During the internship, I worked on multiple research assignments and proof-of-concept projects rather than a single project. Some of the major tasks included:

1. AI Research and Technical Documentation

Prepared documentation on various AI topics, including Generative AI systems, AI infrastructure, AI search techniques, and cloud-based AI solutions.

2. Cloud Computing Research

Studied cloud deployment models and gained hands-on exposure to AWS services such as EC2, S3, IAM, and DynamoDB.

3. n8n Workflow Automation

Built automation workflows using n8n to understand workflow orchestration and business process automation.

4. Node.js and React.js Demonstrations

Developed small demonstration applications to understand web technologies and AI integration concepts.

5. LLM and AI Agent Research

Explored Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), Langflow, Model Context Protocol (MCP), and Reflection Agentic AI patterns using Groq LLM.

3.5 Technologies, Tools, and Software Used

Programming Languages

- JavaScript
- Python

Web Technologies

- Node.js
- React.js
- HTML
- CSS

Cloud Technologies

- Amazon EC2
- Amazon S3
- AWS IAM
- Amazon DynamoDB

AI & Automation Technologies

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- AI Agents
- n8n
- Langflow
- Groq LLM
- Ollama
- Model Context Protocol (MCP)

Development Tools

- Visual Studio Code
- Git
- GitHub
- Google Drive
- Google Meet

3.6 Work Environment and Team Dynamics

The internship was conducted in a mostly remote environment. Weekly tasks were assigned during Google Meet sessions, where mentors introduced new AI concepts, explained the objectives of the assignments, and addressed questions. Day-to-day communication, announcements, and discussions were managed through a WhatsApp group.

Most assignments were completed individually, allowing me to develop independent research and problem-solving skills. For a few activities, I collaborated with a small group of two to three interns to exchange ideas and complete shared tasks. Completed assignments and documentation were submitted through Google Drive, and selected tasks were presented during Google Meet sessions, where mentors provided constructive feedback and suggestions for improvement.

The collaborative environment encouraged continuous learning, knowledge sharing, and professional development while helping me improve communication, teamwork, and presentation skills.

CHAPTER 4

Work and Learning Outcomes

4.1 Detailed Description of Work Done

During the internship at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI), I completed various research-based assignments and practical activities related to Artificial Intelligence, Generative AI, Cloud Computing, and workflow automation. The work was assigned weekly through Google Meet sessions, and I independently researched the topics, prepared technical documentation, developed proof-of-concept demonstrations, and submitted my work through Google Drive.

November 2025 – Introduction to AI Infrastructure and Cloud Computing

The initial phase of the internship focused on understanding the fundamentals of Artificial Intelligence infrastructure and cloud-based AI systems. I studied the differences between cloud-based and on-premises AI deployment models and explored how modern AI applications are deployed in enterprise environments. I also researched AI hardware, private AI systems, and tools such as Ollama for running Large Language Models locally. Technical documentation was prepared based on the learning resources provided by the mentors.

December 2025 – AI Workflow Research and Langflow

During this phase, I explored Langflow, a visual framework used for designing and managing AI workflows. I studied how modular components are connected to build AI pipelines and analyzed workflow configurations represented in JSON format. This helped me understand the architecture of AI applications and the integration of language models into workflow-based systems.

January 2026 – AI Demonstrations and Web Integration

I developed small demonstration projects using Node.js and React.js. One project demonstrated the implementation of the Model Context Protocol (MCP) and integration of the llms.txt file within a React application. Another project explored browser automation using Puppeteer, where repetitive browser tasks were automated to understand AI-assisted automation techniques.

February 2026 – Cloud Computing and AI Tools

This month focused on gaining practical knowledge of Cloud Computing and Amazon Web Services (AWS). I learned the purpose and working of AWS services such as EC2, S3, IAM, and DynamoDB, and understood how cloud platforms support scalable AI applications. I also explored different AI tools and studied their capabilities and real-world use cases.

March 2026 – Large Language Models and AI Technologies

I expanded my knowledge of Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), and prompt engineering. I researched how LLMs process natural language, how AI agents perform autonomous tasks, and how RAG improves the accuracy of AI-generated responses by incorporating external knowledge sources. These concepts provided a strong foundation for understanding modern Generative AI applications.

April 2026 – Reflection Agentic AI Pattern

In the final month of the internship, I implemented the Reflection Agentic AI Pattern using the Groq LLM. This project demonstrated how an AI system can iteratively generate, evaluate, and refine its own responses. I gained practical experience in designing agentic AI workflows, integrating cloud-hosted language models, and understanding self-improving AI systems. This assignment consolidated the knowledge gained throughout the internship and demonstrated the practical application of Generative AI concepts.

4.2 Key Learning in Technical and Soft Skills

Technical Skills

During the internship, I developed practical knowledge in several modern technologies, including:

- Artificial Intelligence and Generative AI concepts
- Large Language Models (LLMs)
- AI Agents and Agentic AI
- Retrieval-Augmented Generation (RAG)
- Workflow automation using n8n
- Cloud Computing fundamentals
- AWS services (EC2, S3, IAM, DynamoDB)
- Node.js and React.js development
- Langflow and Model Context Protocol (MCP)
- Browser automation using Puppeteer
- Technical documentation and research

Soft Skills

The internship also helped me improve several professional skills, including:

- Technical research and analytical thinking
- Independent learning

- Time management
- Communication and presentation skills
- Technical documentation
- Problem-solving
- Team collaboration during group assignments
- Adaptability to new technologies

4.3 Challenges Faced and Solutions Applied

During the internship, I encountered several challenges while learning emerging AI and cloud technologies.

One of the major challenges was understanding advanced concepts such as Large Language Models, AI Agents, Retrieval-Augmented Generation, and cloud architecture, as these topics were new to me. Initially, it was difficult to understand the practical implementation of these technologies.

To overcome these challenges, I regularly referred to the learning resources shared by the mentors, prepared detailed notes, performed additional research using official documentation, and practiced through small hands-on demonstrations. I also actively participated in Google Meet discussions, clarified doubts with mentors, and continuously improved my understanding through self-learning.

Balancing internship assignments with academic responsibilities was another challenge. Proper planning and time management helped me complete the weekly tasks within the given deadlines.

4.4 Contributions to the Organization

During the internship, I contributed by:

- Preparing research-based technical documentation on AI and Cloud Computing topics.
- Developing proof-of-concept demonstrations using Node.js and React.js.
- Creating automation workflows using n8n.
- Exploring AI tools and documenting their features and applications.
- Conducting research on Large Language Models, AI Agents, and RAG.
- Supporting the company's learning and research initiatives related to the AGI Playground platform.
- Maintaining organized documentation and project files for future reference.

4.5 Application of Theoretical Knowledge in Practical Work

The internship provided an opportunity to apply concepts learned during my academic studies to real-world scenarios.

Knowledge of programming, web technologies, databases, and software engineering principles was applied while developing demonstration projects using Node.js and React.js. Concepts of cloud computing were reinforced through practical exposure to AWS services, while Artificial Intelligence concepts were strengthened through research on LLMs, AI Agents, and RAG.

The internship helped bridge the gap between theoretical learning and practical implementation by allowing me to explore modern AI technologies, prepare technical documentation, and understand their applications in enterprise environments.

CHAPTER 5

Industrial Exposure and Skills Development

5.1 Understanding Industry Standards and Practices

The internship at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) provided valuable exposure to the working environment of an AI and Cloud technology startup. I gained an understanding of how research, documentation, experimentation, and continuous learning contribute to the development of AI-based solutions.

The organization followed a structured approach where weekly tasks were assigned through online meetings. Learning resources such as technical articles, research material, and YouTube videos were provided by the mentors, encouraging interns to study independently before implementing or documenting the concepts. This approach helped me understand the importance of self-learning, technical accuracy, and staying updated with rapidly evolving technologies.

I also learned how emerging technologies such as Generative AI, Large Language Models (LLMs), AI Agents, Cloud Computing, and workflow automation are being adopted by industries to build intelligent applications and improve business processes.

5.2 Interaction with Professionals and Mentors

Throughout the internship, I had regular interaction with experienced professionals and mentors through Google Meet sessions. These meetings were conducted on a weekly basis to introduce new topics, explain the objectives of upcoming assignments, discuss industry trends, and resolve technical doubts.

The mentors also reviewed completed assignments, provided constructive feedback, and encouraged independent thinking while solving technical problems. Communication outside the meetings was maintained through a WhatsApp group, where updates, learning resources, and important announcements were shared.

These interactions helped me gain a better understanding of professional work culture and the expectations of the software and AI industry.

5.3 Enhancement of Professional Skills

The internship significantly improved both my technical and professional skills.

Communication Skills

Preparing documentation and presenting selected assignments during online meetings improved my ability to communicate technical concepts in a structured and professional manner.

Teamwork

Although most assignments were completed individually, a few activities involved collaboration with small groups of interns. These tasks helped me understand the importance of coordination, idea sharing, and collaborative problem-solving.

Time Management

Managing internship responsibilities along with academic coursework required proper planning and prioritization. Completing weekly assignments within deadlines strengthened my time management skills.

Research and Analytical Skills

Since the internship was research-oriented, I developed the ability to study new technologies independently, analyze technical concepts, and prepare well-structured documentation based on my findings.

5.4 Networking and Career Growth Insights

The internship provided valuable exposure to professionals working in the fields of Artificial Intelligence and Cloud Computing. Through regular interactions with mentors and fellow interns, I gained insights into current industry trends, emerging technologies, and career opportunities in AI-driven software development.

The experience strengthened my interest in pursuing a career in Artificial Intelligence, Cloud Computing, and Backend Development. It also motivated me to continue learning advanced technologies such as Generative AI, AI Agents, Retrieval-Augmented Generation (RAG), and Large Language Models, which are becoming increasingly important in the software industry.

Overall, the internship helped me build confidence, improve my technical foundation, and better understand the knowledge and skills required for a successful engineering career.

CHAPTER 6

Challenges and Recommendations

6.1 Major Challenges Encountered During Internship

During my internship at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI), I encountered several challenges while learning and working with emerging Artificial Intelligence and Cloud Computing technologies.

One of the primary challenges was understanding advanced concepts such as Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), and workflow automation using n8n. Since these technologies were relatively new to me, it required significant effort to understand their architecture, applications, and practical implementation.

Another challenge was adapting to a research-oriented learning approach. Instead of receiving complete instructions, mentors shared learning resources such as YouTube videos and technical materials. This required me to independently study the topics, analyze the concepts, prepare technical documentation, and perform additional research using official documentation and online resources.

Managing internship responsibilities alongside my academic coursework was also challenging. Completing weekly assignments while attending college lectures, practical sessions, and examinations required effective planning and disciplined time management.

6.2 How They Were Addressed

To overcome these challenges, I adopted a structured learning approach and focused on continuous improvement. I regularly studied the learning resources shared by the mentors and referred to official documentation for technologies such as AWS, Node.js, React.js, and AI-related frameworks. Preparing detailed notes and technical documentation helped me strengthen my understanding of complex topics.

Whenever I faced technical difficulties, I actively participated in the weekly Google Meet sessions to clarify my doubts and understand the concepts more effectively. Feedback provided by the mentors helped me improve the quality of my work and develop a better understanding of industry practices.

To balance internship tasks with my academic responsibilities, I created a weekly schedule and allocated dedicated time for research, documentation, and project development. This approach enabled me to complete assignments within the given deadlines while maintaining my academic performance.

Overall, these experiences improved my independent learning ability, problem-solving skills, and confidence in working with modern AI and cloud technologies.

6.3 Suggestions for Improvement in Future Internships

Based on my experience, I would like to suggest the following improvements that could further enhance the internship program:

- Include more hands-on coding assignments along with research and documentation activities.
- Organize additional live workshops or technical demonstrations on advanced AI and cloud technologies.
- Conduct periodic code review sessions to provide detailed technical feedback.
- Introduce collaborative team-based projects to encourage greater interaction among interns.
- Arrange sessions where experienced professionals share real-world industry case studies and project experiences.
- Provide optional assessments or quizzes after major learning modules to help interns evaluate their understanding and track their progress.

Overall, the internship was highly beneficial and provided valuable exposure to modern technologies, professional work culture, and research-oriented learning.

CHAPTER 7

Conclusion

7.1 Summary of Internship Experience

My six-month internship at INTELLIGENT SYSTEMS PRIVATE LIMITED (EnterpriseSI) was a valuable learning experience that enhanced my understanding of Artificial Intelligence, Generative AI, and Cloud Computing. Throughout the internship, I completed research-based assignments, prepared technical documentation, explored emerging AI technologies, and developed proof-of-concept applications. The internship provided practical exposure to industry-relevant concepts such as Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), workflow automation using n8n, and AWS cloud services.

The structured learning approach, weekly assignments, mentor guidance, and independent research helped me strengthen both my technical and professional skills while gaining insight into current industry practices.

7.2 Key Takeaways and Personal Growth

The internship contributed significantly to my personal and professional development. I improved my ability to learn new technologies independently, conduct technical research, prepare professional documentation, and apply problem-solving skills to real-world scenarios.

Some of the key takeaways from the internship include:

- Understanding the fundamentals of Artificial Intelligence and Generative AI.
- Practical exposure to AWS cloud services, including EC2, S3, IAM, and DynamoDB.
- Hands-on experience with workflow automation using n8n.
- Improved understanding of AI Agents, RAG, LLMs, and cloud-based AI applications.
- Enhanced communication, teamwork, presentation, and time management skills.
- Increased confidence in exploring and adapting to emerging technologies.

The internship also strengthened my analytical thinking and encouraged me to continuously improve my technical knowledge through self-learning.

7.3 How the Internship Influences Future Career Goals

This internship played an important role in shaping my future career goals. It introduced me to advanced technologies such as Artificial Intelligence, Cloud Computing, Generative AI, and intelligent automation, which are transforming the software industry.

The knowledge and practical exposure gained during the internship motivated me to continue developing my skills in software development, backend technologies, cloud computing, and AI-based applications. I aim to build a career where I can contribute to innovative technology solutions while continuously learning and adapting to new advancements in the field.

7.4 Overall Evaluation of the Internship Program

Overall, the internship was highly beneficial and successfully achieved its learning objectives. The program provided a well-balanced combination of technical research, practical learning, independent assignments, and mentor guidance.

The weekly interaction with mentors, continuous learning process, and exposure to modern AI and cloud technologies made the internship an enriching experience. It not only enhanced my technical knowledge but also prepared me for future professional opportunities by improving my confidence, work ethics, and problem-solving abilities.

I am grateful to EnterpriseSI, my mentors, and Vidyalkar Institute of Technology for providing me with this valuable learning opportunity.

CHAPTER 8

References

1. EnterpriseSI. *EnterpriseSI Official Website*. Available: <https://www.enterprisesi.co>
2. Amazon Web Services. *AWS Documentation*. Available: <https://docs.aws.amazon.com>
3. Node.js Foundation. *Node.js Documentation*. Available: <https://nodejs.org/docs>
4. React Team. *React Documentation*. Available: <https://react.dev>
5. n8n. *n8n Documentation*. Available: <https://docs.n8n.io>
6. Langflow Documentation. Available: <https://docs.langflow.org>
7. Groq. *Groq API Documentation*. Available: <https://console.groq.com/docs>
8. Ollama Documentation. Available: <https://ollama.com>
9. DeepLearning.AI. *Agentic AI Learning Resources*.
10. https://drive.google.com/drive/folders/1TDgDUc_OCODvFHKly8frFzdqaTjQFxPI
11. GitHub Repository : <https://github.com/Riya-Kharade/Enterprisesi-ai-cloud-research-internship>

Appendix

Appendix A – Internship Offer Letter

Attach:

- EnterpriseSI Internship Offer Letter

Appendix B – Internship Completion Certificate

Attach:

- EnterpriseSI Internship Completion Certificate

Appendix C – GitHub Repository